

Code-Layout, Readability and Reusability

Date	29-09-2026
Team ID	SWTID-2026-1297
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S . N o	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation is followed throughout the code.
2	Proper File Structure	Files and folders are logically organized	Yes	Files and folders are logically organized into config, controllers, middleware, models, routes, services and utils.
3	Meaningful variable names	Meaningful Names Variables reflect their purpose	Yes	Variables and functions use meaning
4	Function / Method Names	Functions are descriptively named	Yes	Functions are descriptively named according to their functionality.

5	Code Comments		Inline and block comments explain logic	Partial	Comments are used where required; additional comments can be added for complex logic.
6	Modular Design		Code is split into reusable functions/modules	Yes	The application is divided into reusable modules such as controllers, routes, services and middleware.
7	No Redundant Code		Duplicate or unused code is removed	Yes	The code is organized to avoid unnecessary duplicate code.
8	Error Handling		Exceptions and errors are handled gracefully	Yes	Errors are handled using error-handling middleware and appropriate responses.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Tech	Where Reused	Reusability Level (High / Medium / Low)
1	authMiddleware.js	Node.js / Express.js	Protected API routes	High

2	errorMiddleware.js	Node.js / Express.js	API endpoints	High
3	validationMiddleware.js	Node.js / Express.js	Request validation	High
4	geminiService.js	Node.js / Google Gemini AI	AI answer generation	High

5	helpers.js	JavaScript / Node.js	Common utility operations	Medium
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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Files and folders are logically organized into separate modules.
Readability	4	Meaningful names and a clear structure are used throughout the project.
Reusability	4	Middleware, services and utility modules are designed for reuse.
Documentation / Comments	3	Basic comments and documentation are provided; some areas can be documented further.
Overall Score	4	The project has a well-structured, readable and reusable codebase with scope for further documentation improvements.